

Segment Arithmetic

The Sum of Line Segments

Definition (the sum of line segments)

Given congruence equivalence classes of line segments a, b , choose points A, B such that the segment AB represents class a , then, on line AB , choose point C such that $A * B * C$ and the segment BC represents class b . We define the *sum* $a + b$ to be represented by the segment AC .



Addition of Line Segments

Proposition

In any Hilbert plane, *addition* of line segment classes has the following properties:

1. $a + b$ is well-defined, i.e., different choices of A, B, C in the definition will give rise to congruent segments.
2. $a + b = b + a$, i.e., the corresponding line segments are congruent.
3. $(a + b) + c = a + (b + c)$.
4. Given any two classes a, b , one and only one of the following holds: (i) $a = b$. (ii) There is a class c such that $a + c = b$. (iii) There is a class d such that $a = b + d$.



Addition of Line Segments

Proof. (1) Take $A'B'$ to be another representative of class a . Choose C' on line $A'B'$ such that $B'C'$ represents class b with $A' * B' * C'$. Then, by (C3), $AC \cong A'C'$. (2) Let AB represent class a . Choose C on line AB such that BC represents class b with $A * B * C$. Hence, AC represents $a + b$. Now, take DE to represent class b . Choose F on line DE such that EF represents class a with $D * E * F$. Hence, DF represents $b + a$. However, $AB \cong FE$ and $BC \cong ED$, so $AC \cong FD$ by (C3). Q.E.D.

The Product of Line Segments

Definition (the product of line segments)

Given congruence equivalence classes of line segments a, b , make a right-triangle ABC with $AB \in 1$ and $BC \in a$ where the right-angle is at B , take α to be the angle BAC . Then, make a right-triangle DEF with $DE \in b$, right-angle at E , and α at D . We define the *product* ab to be represented by the segment EF .



Multiplication of Line Segments

Proposition

In any Hilbert plane with (P), *multiplication* of line segment classes has the following properties:

1. ab is well-defined.
2. $a \cdot 1 = a$ for all a .
3. $ab = ba$ for all a, b .
4. $a(bc) = (ab)c$ for all a, b, c .
5. For any a , there is a unique b such that $ab = 1$.
6. $a(b + c) = ab + ac$ for all a, b, c .



Multiplication of Line Segments

Proof. (1) Take $A'B'C'$ to be another right-triangle with $A'B' \in 1$ and $B'C' \in a$ where the right-angle is at B' . Then, by SAS, $A'B'C'$ is congruent to ABC . Hence, we get the congruent angle α at angle $B'A'C'$. If $D'E'F'$ is another right-triangle with angle α at angle $E'D'F'$, side b , and right-angle at E' , then, by ASA, $D'E'F'$ is congruent to DEF . Hence, we get the congruent side ab at segment $E'F'$. (2) Take right-triangle DEF to have angle α , side $b = 1$, and right-angle at E . By ASA, DEF is congruent to ABC , so $a \cdot 1 = a$. Q.E.D.

Similar Triangles

Definition (similarity of triangles)

Two triangles ABC and $A'B'C'$ are *similar* if the three angles of one are respectively equal to the three angles of the other, and the corresponding sides are proportional, i.e.,

$$\frac{a}{a'} = \frac{b}{b'} = \frac{c}{c'}.$$



AAA Similarity

Proposition (AAA similarity)

If two triangles ABC and $A'B'C'$ have their three angles respectively equal, then the two triangles are similar.



AAA Similarity

Proof. ...

Q.E.D.

SSS Similarity

Proposition

In any triangle ABC , let $B'C'$ be drawn parallel to BC . Then, the sides AB and AC are proportional to AB' and AC' . Conversely, if the sides are divided by points B', D such that AB, AC are proportional to AB', AD , then $B'D$ is parallel to BC .



SSS Similarity

Proposition (SSS similarity)

If two triangles ABC and $A'B'C'$ have their three sides respectively proportional to each other, then the two triangles are similar.



SAS Similarity

Proposition (SAS similarity)

If two triangles ABC and $A'B'C'$ have the angles at A and A' equal and the two sides AB, AC are proportional to the two sides $A'B', A'C'$ respectively, then the two triangles are similar.



Euclid's Theory of Similar Triangles

Theorem

In any Hilbert plane with (P), the results of Euclid's theory of similar triangles (VI.2)-(VI.13) all hold.



The Pythagorean Theorem

Proposition (the Pythagorean theorem)

If ABC is a right-triangle with legs a, b and hypotenuse c , then $a^2 + b^2 = c^2$ in the field F of segment arithmetic.

